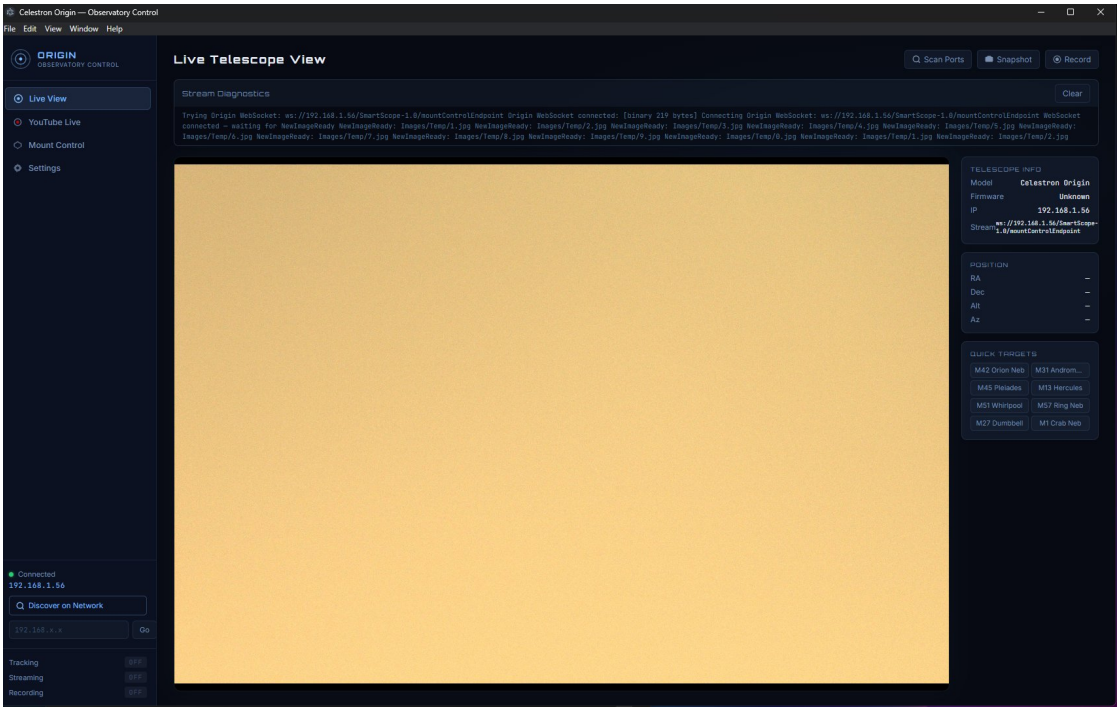
# CELESTRON ORIGIN

## Observatory Control

*Complete User Manual*




Windows Desktop Application | Electron + Node.js | Version 2.1
Compatible with Celestron Origin Intelligent Home Observatory

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# 1 — Getting Started

## 1.1 System Requirements

| | |
|---|---|
| **OS** | Windows 10 or later (64-bit) |
| **RAM** | 4 GB minimum, 8 GB recommended |
| **Disk** | 500 MB free space |
| **Network** | Must be on same Wi-Fi network as the telescope |
| **FFmpeg** | Included automatically during installation |
| **Node.js** | Included automatically during installation |

## 1.2 Installation

The app ships as a ZIP file. No administrator rights are required.

- Extract **CelestronOriginControl-Installer.zip** to any folder
- Double-click **Install.bat** to launch the installer
- The installer downloads Node.js and FFmpeg automatically if needed
- A desktop shortcut is created automatically
- To uninstall, run **Uninstall.bat**

■ The installer does not require administrator privileges. All files install to your user folder.

## 1.3 Connecting to Your Telescope

Power on your Celestron Origin and connect your Windows PC to the same Wi-Fi network as the telescope. Then:

- **Discover on Network** — click this button in the left sidebar to auto-scan for the telescope (5–10 seconds)
- **Manual IP** — type the telescope IP directly (typically 192.168.1.56) and click Go

Once connected the sidebar shows:

| | |
|---|---|
| **Green dot + "Connected"** | Telescope is connected and responding |
| **IP Address** | The telescope's current network address |
| **Tracking badge** | Shows ON when sidereal tracking is active |
| **Streaming badge** | Shows LIVE when a stream is in progress |
| **Recording badge** | Shows ON when local recording is active |
| **Image Mode badge** | Shows LIVE or STACKED depending on current image mode |

# 2 — Live View

## 2.1 Live View Interface

The Live View tab shows a real-time feed from the telescope camera at ~1 fps via the SmartScope WebSocket. The full 3056×2048 JPEG is transmitted and displayed scaled to the window.

| | |
|---|---|
| **Model** | Telescope model (Celestron Origin) |
| **Firmware** | Installed firmware version |
| **IP** | Active network address |
| **RA / Dec** | Current right ascension and declination |
| **Alt / Az** | Current altitude and azimuth in degrees |

## 2.2 Quick Controls

| | |
|---|---|
| **Scan Ports** | Opens/closes Stream Diagnostics — raw WebSocket log for debugging |
| **Snapshot** | Saves a full-resolution JPEG to your Documents folder |
| **Record** | Starts/stops local MP4 recording to your Documents folder |

## 2.3 Tonight's Best Targets

A live list of deep sky objects well-positioned from your location, ranked by altitude. Location is read automatically from the telescope GPS. Click any target to immediately slew to it.

| | |
|---|---|
| **GREAT badge** | Object above 50° — excellent conditions |
| **GOOD badge** | Object between 20°–50° — acceptable conditions |

## 2.4 Viewing Conditions

Fetches live weather from **Open-Meteo** and calculates an overall observing score (0–100).

| | |
|---|---|
| **80–100 Excellent** | Clear skies, low humidity, calm winds |
| **60–79 Good** | Mostly clear with minor limitations |
| **40–59 Fair** | Some cloud or wind |
| **20–39 Poor** | Significant cloud cover |
| **0–19 Bad** | Overcast, rain or fog |

| | |
|---|---|
| **Cloud Cover** | Primary factor — clear sky scores maximum points |
| **Precipitation** | Any rain/snow immediately deducts 60 points |
| **Dew Risk** | Temperature vs dew point gap — below 3°F = dew warning |
| **Wind Speed** | Calm (≤5 mph) ideal; strong winds cause blur |
| **Visibility** | Atmospheric transparency — haze and fog reduce score |

■ Weather data refreshes every 15 minutes. Use the ■ button to refresh manually.

## 2.5 Image Stacking

When the telescope is initialized and tracking, you can enable image stacking to build up a deep-sky image from multiple exposures. The live view canvas automatically switches from showing single live frames to showing the accumulating stacked image.

### Stacking Controls (in the Telescope tab)

| | |
|---|---|
| ■ **Start Stacking** | Sends SetEnableAuto to LiveStream — telescope begins stacking exposures |
| ■ **Stop Stacking** | Returns to single-frame live view mode |
| **Stacking status** | Shows current state: stacking active / not stacking |

### Stacking Overlay

When stacking is active a compact overlay appears at the top of the live view showing:

| | |
|---|---|
| **Target name** | Name of the celestial object being imaged (from most recent GoTo) |
| **Frames** | Number of stacked exposures received so far |
| **Integration** | Total accumulated exposure time (e.g. 1m 30s) |

■ *Stacking requires a successful nighttime initialization. Demo Mode initialization does not enable stacking.*

## 2.6 Snapshot, Record, and Scan Ports

### Snapshot

Captures the current frame at full resolution (3056×2048). Saved as Origin_YYYYMMDD_HHMMSS.jpg in Documents.

### Record

Starts an MP4 recording via FFmpeg. Click again to stop — file is finalized and saved to Documents with a timestamp.

■ *Recording and streaming can run simultaneously.*

### Stream Diagnostics

Raw WebSocket log. Useful if live view is not appearing — shows connection attempts, NewImageReady messages, and errors.

# 3 — Live Streaming

The Live Stream tab streams to YouTube or Twitch via RTMP using FFmpeg H.264 encoding, with optional SomaFM background music.

## 3.1 YouTube Streaming Setup

YouTube streaming uses a stream key — no OAuth or developer setup required.

### Step 1 — Open YouTube Studio

- Go to **studio.youtube.com**
- Click **Stream** in the left sidebar
- If prompted, click **Now** then select **Streaming software**

### Step 2 — Copy Your Stream Key

- You will see the Live Control Room page
- At the top of the preview area, find **"Stream key: Default stream"**
- Click the **copy icon** (two squares) next to it — this copies your stream key
- The Stream URL shown is: **rtmp://a.rtmp.youtube.com/live2**

### Step 3 — Paste into App and Go Live

- In the app's Live Stream tab, make sure **YouTube** is selected
- Paste the stream key into the **Stream Key** field
- Keep the YouTube Studio page open in your browser
- Click **Go Live on YouTube** in the app
- After ~8 seconds the button changes to **Stop Stream** confirming the connection
- In YouTube Studio you will see your telescope feed appear in the preview
- Click **Go Live** in YouTube Studio to make the stream public to viewers

■ *The stream key is intentionally hidden by YouTube. The copy icon next to "Stream key: Default stream" is the only way to get it — there is no visible text to select.*

## 3.2 Twitch Streaming

Twitch requires only a stream key — no developer project needed.

- Click the **Twitch** button at the top of the Live Stream tab
- Go to **dashboard.twitch.tv** → **Settings** → **Stream** to find your stream key
- Paste the key into the Stream Key field
- Click **Go Live on Twitch**

| | |
|---|---|
| **RTMP Server** | rtmp://live.twitch.tv/app/ |

| | |
|---|---|
| **Key Format** | live_XXXXXXXXXX_XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX |
| **Encoder** | FFmpeg libx264 (H.264) |
| **Audio** | AAC stereo |

## 3.3 Background Music — SomaFM Radio

Add royalty-free internet radio to your stream via SomaFM. Because it is a live stream it never ends — no interruptions while live.

- Toggle the **Enable** switch under Background Music
- Click any station from the grid to select it
- Adjust the **Volume** slider (default 40%)
- You can switch stations or adjust volume live — the main stream never drops

| | |
|---|---|
| ■ **Space Station** | Ambient / Space — ideal for astronomy streams |
| ■ **Deep Space One** | Deep Ambient — slow, atmospheric |
| ■ **Drone Zone** | Dark Ambient — minimal and hypnotic |
| ■ **Groove Salad** | Ambient / Electronic — mellow beats |
| ■ **Lush** | Lush Downtempo — warm and melodic |
| ■ **Cliqhop** | IDM / Electronic |
| ■ **Sonic Universe** | Jazz / Fusion |
| ■ **Ill Street Blues** | Blues / Jazz |
| ■ **Beat Blender** | Downtempo / Hip-Hop |
| ■ **Indie Pop** | Indie / Pop |
| ■ **Folk Forward** | Folk / Indie |
| ■ **Thistle Radio** | Celtic / Folk |

## 3.4 Quality Settings

| | |
|---|---|
| **480p** | 854×480 @ 30fps · 2500 kbps video · 96 kbps audio |
| **720p HD** | 1280×720 @ 30fps · 4500 kbps video · 128 kbps audio (recommended) |
| **1080p FHD** | 1920×1080 @ 30fps · 6000 kbps video · 192 kbps audio |

■ *The telescope delivers ~1 fps. FFmpeg interpolates to reach the target frame rate.*

# 4 — Mount Control

## 4.1 D-Pad Slew Controls

Hold a direction button to slew. Release to stop. The center ■ button sends an immediate stop.

| | |
|---|---|
| ▲ North | Increase altitude |
| ▼ South | Decrease altitude |
| ■ West | Move west |
| ■ East | Move east |
| ■ Stop | Immediately halt all slew motion |

### Speed Control

| | |
|---|---|
| 1× | Rate 1 — very slow, precise fine adjustments |
| 3× | Rate 3 — slow centering |
| 6× | Rate 6 — medium speed (default) |
| 9× | Rate 9 — fast large movements |

### Tracking & Parking

| | |
|---|---|
| Sidereal Tracking ON | Enables sidereal tracking — mount follows stars |
| Stop Tracking | Disables tracking |
| Park Mount | Sends telescope to home position |
| ■ EMERGENCY STOP | Immediately halts all motion and disables tracking |

## 4.2 Go-To Coordinates

- **RA (HH MM SS)** — right ascension in hours, minutes, seconds
- **Dec (DD MM SS)** — declination with sign (e.g. -05 23 28)
- Click **Slew to Coordinates** to begin the GoTo

■ Decimal degrees also work — e.g. 83.8 for RA, -5.4 for Dec.

## 4.3 Tonight's Best Targets

The same Tonight's Best list from Live View appears here. Click any target to slew immediately.

## 4.4 Initialize Telescope

Initialization is required before the telescope can perform GoTo slews or image stacking. It runs a full startup sequence: battery check, hardware calibration, dark frame generation, flat frame generation, focus test, and star alignment.

| | |
|---|---|
| ■ **Initialize Telescope** | Full initialization — telescope must be outdoors at night with lens cap on |
| ■ **Demo Mode Initialize** | Skips sun-above-horizon check for daytime testing. Uses FakeInitialize:true. Runs the sequence but without real star alignment. |

The status box shows the current phase in real time:

| | |
|---|---|
| ■ **Sending command…** | RunInitialize command sent to telescope |
| ■ **Initialization running…** | Telescope is executing the startup sequence |
| ■ **Sun is above horizon…** | Normal daytime error — use Demo Mode Initialize for testing |
| ✓ **Initialization complete** | Telescope is ready for GoTo and stacking |

■ *A full nighttime initialization takes 2–3 minutes. Do not disconnect the telescope during this process.*

■ All commands (slew, initialize, stacking) now reuse the main WebSocket connection, which improves reliability and response time.

## 4.5 Focus and Camera

| | |
|---|---|
| **In / Out buttons** | Move focuser by the selected step size |
| **Step slider** | Steps per button press (1–10) |
| **Gain** | Camera ISO equivalent |
| **Exposure** | Exposure time in milliseconds |
| **Brightness** | Post-capture brightness adjustment |

# 5 — Sky Chart

The Sky Chart tab shows a real-time interactive map of the sky based on your current location and time. Objects are calculated using local sidereal time and your telescope's GPS coordinates. Use it to explore what's visible and slew to targets directly.

## 5.1 Navigating the Sky Chart

| | |
|---|---|
| **Drag left / right** | Rotate the view around the horizon (360° azimuth pan) |
| **Drag up / down** | Tilt the view up toward the zenith or down toward the horizon |
| **Scroll wheel up** | Zoom in — narrow field of view, objects spread out |
| **Scroll wheel down** | Zoom out — wide field of view, see more of the sky |
| **■ reset (bottom right)** | Resets rotation, tilt, and zoom to default |

The compass bar below the chart always shows the current azimuth centre, altitude centre, and field of view in degrees (e.g. **Az 180° · Alt 30° · 90° view**).

| | |
|---|---|
| **Altitude grid lines** | Horizontal lines at 0°, 15°, 30°, 45°, 60°, 75°, 90° |
| **Azimuth lines** | Vertical lines at N/NE/E/SE/S/SW/W/NW with labels at bottom |
| **Horizon line** | Thick line at 0° — objects below are dimmed |
| **Cardinal labels** | N/E/S/W shown in bright blue; NE/SE/SW/NW dimmer |

■ Start with a wide zoom (scroll down) to orient yourself, then scroll in to zoom on a specific area of sky before selecting a target.

## 5.2 Selecting & Slewing to Objects

Hover over any object on the canvas to see its name highlighted. Click to select it:

| | |
|---|---|
| **Selection rings** | Two concentric rings appear around the selected object |
| **View pans** | The chart automatically centres on the selected object |
| **Info card** | Appears below the chart showing altitude, azimuth, and type |
| **■ Slew Telescope Here** | Sends GoTo command to the telescope (requires connection) |
| **✕ close** | Dismisses the info card and deselects the object |

You can also click any row in the object list below the chart to select it — the chart will pan to show it and the info card will appear.

■ *Clicking an object only selects it and shows information. It does NOT automatically slew the telescope. You must click "Slew Telescope Here" to initiate a GoTo.*

### Object Types and Colours

| | |
|---|---|
| **Gold ●** | Stars — size indicates brightness (larger = brighter) |
| **Cyan ■** | Nebulae — emission/reflection nebulae |

| | |
|---|---|
| **Purple ■** | Galaxies — spiral and elliptical galaxies |
| **Teal ■** | Clusters — open and globular clusters |
| **Orange ●** | Planets — approximate positions |

## 5.3 Search

Type in the search bar at the top of the Sky Chart tab to filter both the canvas and the object list:

- Search by name — e.g. "Orion", "M42", "Andromeda"
- Search by type — e.g. "nebula", "galaxy", "cluster", "planet"
- Results update instantly as you type
- The object list below shows only matching objects sorted by altitude
- Clear the search box to return to the full sky view

■ The Sky Chart uses your telescope GPS coordinates automatically when connected. If not connected, it falls back to your saved location from Settings.

# 6 — Telescope Settings

The Telescope tab displays live data from the SmartScope API. Refresh with the ■ button.

■ *The telescope must be connected for this tab to display data.*

## 6.1 System Information

| | |
|---|---|
| **Model** | Telescope model name |
| **Firmware** | Current firmware version (e.g. 1.3.5330) |
| **Serial Number** | Unique hardware serial number |
| **Camera Model** | Sensor model identifier |
| **Update Channel** | "stable" or "beta" firmware track |
| **Update Available** | Whether a newer firmware is available |

## 6.2 Camera Settings

| | |
|---|---|
| **Resolution** | Full sensor resolution (3056 × 2048) |
| **Pixel Size** | Physical pixel size (2.4 µm) |
| **ISO Range** | Min and max ISO sensitivity |
| **Exposure Range** | Min and max exposure times in seconds |
| **White Balance R/G/B** | Current colour balance values |

## 6.3 Mount Configuration

| | |
|---|---|
| **Battery Level** | HIGH / MEDIUM / LOW — colour coded |
| **Battery Voltage** | Actual voltage in volts (nominal ~9.9V) |
| **Charger Status** | CHARGING / DISCHARGING / CHARGED |
| **Cordwrap** | Prevents cable tangling during rotation |
| **Alt / Azm Backlash** | Axis backlash compensation values |

■ Battery voltage below 9.5V means charge soon. Below 9.0V the telescope may shut down.

## 6.4 Focuser Advanced Settings

| | |
|---|---|
| **Position** | Current absolute focuser position (motor steps) |
| **Auto-focus after GoTo** | Runs auto-focus after each GoTo slew |
| **Temp shift auto-focus** | Triggers auto-focus on temperature change |
| **Temp threshold** | Temperature change to trigger auto-focus (default 5°C) |

## 6.5 Environment & Dew Heater

| | |
|---|---|
| **Ambient Temperature** | External temperature near the tube (°C) |
| **Camera Temperature** | Temperature at the sensor (°C) |
| **Humidity** | Relative humidity — dew risk above ~80% |
| **Dew Heater Mode** | Auto (managed automatically) or Manual (fixed level) |
| **Aggression** | How aggressively auto mode responds to dew risk (1–10) |

## 6.6 Network & LED Ring

| | |
|---|---|
| **Telescope IP** | Current IP address on your network |
| **Internet** | Whether telescope has internet access |
| **Force Direct Connect** | Always create own hotspot instead of joining network |
| **LED Ring Brightness** | 0.0 = off, 1.0 = full brightness |

# 7 — App Settings

## 7.1 Observer Location

Your location is set **automatically** — the telescope broadcasts GPS coordinates every second. The app reads this within 1–2 seconds of connecting. As a fallback, enter coordinates manually:

| | |
|---|---|
| **Latitude** | Decimal degrees — North is positive (e.g. 42.58) |
| **Longitude** | Decimal degrees — West is negative (e.g. -70.77) |
| **Save Location** | Persists to disk for use before the telescope connects |

## 7.2 Connection Settings

| | |
|---|---|
| **Default IP** | Pre-fills the manual IP field on app start |
| **Timeout (ms)** | How long to wait for responses (default 5000ms) |
| **Auto-reconnect** | Re-connects automatically if the telescope connection drops |

## 7.3 FFmpeg

FFmpeg is the video encoding engine for streaming and recording. It installs automatically.

| | |
|---|---|
| **FFmpeg status** | Shows installed version or not-found status |
| **Default Recording Quality** | Default quality for new recordings |
| **Open FFmpeg Download Page** | Opens ffmpeg.org if manual installation needed |

■ *If FFmpeg shows as not found, re-run Install.bat to repair the installation.*

# 8 — Troubleshooting

## 8.1 No Live View (blank or frozen canvas)

- Confirm the app shows "Connected" in the sidebar
- Check that the Celestron mobile app is not connected simultaneously — the telescope may only allow one WebSocket client
- Try disconnecting and reconnecting using the Discover button
- The first frame takes 2–3 seconds — wait briefly before assuming it is broken

## 8.2 Stream Stuck on "Starting…"

- The stream is likely working on YouTube Studio but the app UI has not updated
- After ~8 seconds the button should change to Stop Stream automatically
- If still stuck, check the FFmpeg log in the stream tab for connection errors
- Ensure the YouTube Studio Streaming software page is open before clicking Go Live
- The stream key must be pasted in the app — click the copy icon next to "Stream key: Default stream" in YouTube Studio
- To stop streaming: click Stop Stream — the app resets immediately regardless of FFmpeg state

## 8.3 Initialize Shows Error

| | |
|---|---|
| **"Sun is above horizon"** | Normal daytime error. Use ■ Demo Mode Initialize for testing during the day. |
| **"No status received"** | Was shown in older versions. Now resolved — commands reuse main WebSocket. |
| **"Initialization timed out"** | Initialization ran but took over 5 minutes. Check telescope is responding. |
| **Latitude/Longitude missing** | Connect telescope first so GPS can be read, then initialize. |

■ Always connect to the telescope and wait 2–3 seconds for GPS to be read before clicking Initialize.

## 8.4 D-Pad Controls Not Working

- The telescope must be initialized before Slew commands are accepted
- All commands now reuse the main WebSocket — no separate connection needed
- If the telescope has disconnected, re-connect and try again

## 8.5 Tonight's Best Targets Not Showing

- GPS is read from the telescope automatically — wait 1–2 seconds after connecting
- As a fallback, go to **Settings** → **Observer Location** and enter coordinates manually

- Objects may all be below the 20° threshold during daytime — check back after dark

## 8.6 Viewing Conditions Not Loading

- Requires internet access to reach api.open-meteo.com
- If on the telescope's own Wi-Fi hotspot, that network may not have internet
- Use the ■ Refresh button to retry

## 8.7 Radio Station Not Playing / No Audio

- Select a station in the grid before clicking Go Live
- SomaFM requires an active internet connection
- Volume at 0% mutes audio — check the Volume slider
- Try a different station if one is temporarily down

# Quick Reference

## Navigation Tabs (left to right)

| | |
|---|---|
| **Live View** | Real-time telescope feed, stacking overlay, snapshot/record |
| **Live Stream** | YouTube / Twitch streaming with SomaFM radio |
| **Mount Control** | D-pad, GoTo, Initialize, tracking, focus |
| **Sky Chart** | Interactive 360° night sky map with zoom and GoTo |
| **Telescope** | Live telescope settings and sensor data |
| **Settings** | Location, connection, FFmpeg configuration |
| **About** | App info and Ko-fi support link |

## Sky Chart Controls

| | |
|---|---|
| **Drag left/right** | Rotate azimuth 360° |
| **Drag up/down** | Tilt altitude view |
| **Scroll up** | Zoom in (narrower field) |
| **Scroll down** | Zoom out (wider field) |
| **Click object** | Select — shows info card |
| **■ Slew Telescope Here** | GoTo selected object |
| **■ reset** | Reset zoom, rotation, tilt |

## Confirmed Telescope API Commands

All commands use: **ws://[IP]/SmartScope-1.0/mountControlEndpoint**

| | |
|---|---|
| **Slew** | Mount directional movement (AltRate, AzmRate: -9 to 9) |
| **GotoRaDec** | GoTo equatorial coordinates |
| **RunInitialize** | Initialize mount (Date, Time, TimeZone, Lat, Lon, FakeInitialize) |
| **SetEnableAuto → LiveStream** | Start image stacking |
| **SetEnableManual → LiveStream** | Stop stacking / manual mode |
| **GetVersion** | Fetch firmware version |
| **GetStatus (TaskController)** | Init phase: State + Stage fields |
| **GetStatus (Mount)** | GPS, battery, tracking, position |

## RTMP Endpoints

| | |
|---|---|
| **YouTube** | rtmp://a.rtmp.youtube.com/live2/[STREAM_KEY] |
| **Twitch** | rtmp://live.twitch.tv/app/[STREAM_KEY] |

**Celestron Origin Observatory Control** is an unofficial third-party desktop companion. Not affiliated with or supported by Celestron LLC.
Protocol reverse-engineered from network packet analysis for interoperability purposes.